

Topic 2: Structure and function of living organisms – Cells and organisation

Students should:		Guidance <i>iLowerSecondary Curriculum reference</i>
2.1	know the structure of a typical animal cell	membrane surrounds cell, cell contains cytoplasm, mitochondria and nucleus <i>Year 7: Structure and function of living organisms – Cells and organisation</i>
2.2	know the similarities and differences between plant and animal cells	know that plant cells have a cell wall, chloroplasts and a permanent vacuole in addition to the parts an animal cell has <i>Year 7: Structure and function of living organisms – Cells and organisation</i>
2.3	know how to identify the cell wall, cell membrane, cytoplasm, nucleus, permanent vacuole, mitochondria and chloroplasts in a range of familiar and less familiar animal and plant cells	<i>Year 7: Structure and function of living organisms – Cells and organisation</i>
2.4	know the functions of the cell wall, cell membrane, cytoplasm, nucleus, permanent vacuole, mitochondria and chloroplasts	<i>Year 7: Structure and function of living organisms – Cells and organisation</i>
2.5	know the hierarchical organisation of multicellular organisms from cells to tissues to organs to organ systems to organisms	<i>Year 7: Structure and function of living organisms – Cells and organisation</i>
2.6	know the major organs and organ systems of the human body and their functions	lungs, kidney, skin, brain, heart, liver, stomach (small and large intestine), nervous system, skeleton and muscles <i>Year 7: Structure and function of living organisms – Cells and organisation</i>
2.7	know the similarities and differences between generalised plant, animal and bacterial cells	similarities – plant, animal and bacterial cells all possess cell membranes and cytoplasm differences – bacteria cells possess no nucleus and no mitochondria and are much smaller <i>Year 8: Structure and function of living organisms – Cells and organisation</i>